

MTSST: Inherently Interpretable Method for Prompt Classification of Multivariate Time Series

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Abstract. Time series classification (TSC) is a critical task in domains such as healthcare and finance, where timely and explainable decisions are often required. While deep learning models achieve high accuracy, they typically lack interpretability and require complete data sequences. In this paper, we introduce the Multivariate Time Series Step Tree (MTSST), a novel, inherently interpretable decision-tree-based method designed for the prompt classification of multivariate time series. MTSST adopts a progressive, step-wise evaluation strategy, dynamically adapting its observation window to classify partial time series as soon as sufficient discriminatory information is available. Our method leverages a "witness time series" selection mechanism to maximize information gain at each node, providing transparent, human-readable decision paths. We extend the core TSST model to multivariate data using a random forest ensemble that trains on specific dimensions, capturing cross-channel dependencies while maintaining the interpretability of individual trees. Comprehensive experiments on univariate and multivariate datasets demonstrate that MTSST achieves state-of-the-art performance while significantly reducing decision latency, making it highly suitable for real-time, high-stakes applications.

Keywords: Time Series Classification · Prompt Classification · Interpretability · Multivariate Time Series · Decision Trees.

1 Introduction

In many real-world scenarios, specially in critical domains like healthcare, finance, and industrial monitoring, decision-making often relies on time series (*TS*) data. A key requirement in these fields is the ability to classify and make predictions based on evolving data, often before the complete sequence is observed [6,10]. For instance, in healthcare, continuous monitoring of vital signs can inform immediate patient care decisions. The challenge lies in developing models that are not only accurate but also *prompt*—minimizing the time to decision—and *interpretable*, providing transparent reasoning for their outputs [2].

Traditional time series classification (TSC) approaches have evolved significantly, but a gap remains in simultaneously addressing dynamic early classification and interpretability. Deep learning models, while powerful, often function

as black boxes [6]. Conversely, standard decision trees offer interpretability but are typically designed for static, complete datasets. Existing early classification methods [3,11] often rely on fixed or optimized observation windows rather than dynamically adapting to the specific temporal context of each instance.

To bridge this gap, we propose the **Multivariate Time Series Step Tree (MTSST)**. Building upon the Time Series Step Tree (TSST) concept for univariate data, MTSST is designed explicitly for the prompt classification of multivariate time series. It employs a progressive approach, evaluating *TS* data in "steps" and dynamically deciding whether to make a classification or observe more data. Central to our approach is the novel **Witness-TS** mechanism, which selects the most discriminative subsequence from the training data to guide the splitting process at each tree node.

Our contributions are as follows:

- We introduce **MTSST**, a novel interpretable method for multivariate time series classification that enables prompt decision-making with dynamic observation windows.
- We propose a **Random Forest extension** of the TSST model that effectively handles multivariate data by training trees on specific dimensions, capturing diverse patterns across channels.
- We demonstrate through extensive experiments that MTSST achieves robust accuracy comparable to state-of-the-art methods while enabling significant reductions in observation time, and we provide visual insights into its interpretable decision-making process.

The remainder of this paper is organized as follows. Section 2 reviews related work. Section 3 details the MTSST model, including the Witness-TS selection and the multivariate random forest extension. Section 4 presents our experimental evaluation on univariate and multivariate datasets. Finally, Section 5 concludes the paper.

2 Related Work

Time Series Classification (TSC) has seen extensive research, with approaches ranging from distance-based methods like 1-NN DTW to sophisticated deep learning architectures. However, for applications requiring timely decisions, *early classification* has emerged as a key subfield [9,4].

Early Classification. Algorithms such as ECTS [7] and TEASER [3] aim to classify streams without waiting for the full sequence. TEASER mines "early shapelets" and uses a greedy search to balance accuracy and earliness. ECONOMY [11] frames the problem as a cost-optimization task. While effective, these methods often rely on fixed or optimized observation windows and may lack deep interpretability.

Interpretability. Decision trees (DTs) remain a gold standard for interpretability [2] due to their rule-based structure. However, standard DTs often struggle with the dynamic nature of early TSC. Recent works like CALIMERA

[1] and Stop&Hop [5] improve performance but often trade off transparency for accuracy, using complex latent representations.

The Gap. There exists a need for a model that is inherently interpretable and can dynamically adapt its observation window on a *case-by-case* basis. MTSST fills this gap by utilizing a decision tree structure where each node represents a specific "step" in the time series, allowing for transparent, witness-guided splitting and naturally enabling prompt classification.

3 Methodology: The MTSST Model

The core of our proposal is the Time Series Step Tree (TSST), which we extend to the multivariate domain (MTSST). The model operates by evaluating *TS* data in progressive steps. At each node of the tree, the model decides whether it has sufficient information to classify the instance or if it needs to observe more "steps" (features) of the time series.

3.1 Distance Calculation and Catch22

To determine the optimal split at each node, TSST dynamically selects the distance metric that yields the highest information gain. We utilize a range of distances including Euclidean, DTW, and correlation-based metrics. Additionally, we incorporate *Catch22* [8], a set of 22 computationally efficient time-series features. When selected, the model computes a pairwise distance matrix based on these features, reducing computational cost while maintaining discriminative power.

3.2 Witness Time Series (Witness-TS)

A key innovation of TSST is the **Witness-TS**. Instead of splitting based on arbitrary thresholds, each node in the TSST tree is associated with a specific subsequence from the training set, called the Witness-TS. The splitting process works as follows:

1. For a given node, we evaluate potential Witness-TS candidates from the training data.
2. For each candidate, we compute the distance between it and all other samples up to the current "step" (observation window).
3. We identify the optimal distance threshold that splits the data into two branches (True/False) to maximize **Information Gain** (reduction in entropy).
4. The candidate yielding the highest gain is selected as the Witness-TS for that node.

This mechanism ensures that every split is grounded in a concrete, interpretable reference pattern.

3.3 TSST Train and Predict

The training algorithm constructs the tree recursively. It starts with a small observation window (controlled by a ‘stepsize’ parameter). If a pure classification cannot be made, the tree grows, potentially increasing the observation window in child nodes. The prediction algorithm follows this structure: it starts at the root, compares the incoming partial TS to the Witness-TS, and traverses the tree. The traversal stops at a leaf node, providing a classification decision. This naturally handles "promptness": simple instances may reach a leaf node after only a few steps, while complex instances traverse deeper, determining their own necessary observation length.

3.4 Multivariate Extension (MTSST) with Random Forest

To handle multivariate time series, we propose a Random Forest (RF) extension of TSST.

- **Channel Independence:** Unlike approaches that flatten multivariate data, our RF approach treats each dimension (channel) as a separate input source.
- **Tree-wise Randomization:** Each tree in the forest is randomly assigned to a specific dimension of the multivariate dataset during training.
- **Ensemble Voting:** During prediction, each tree casts a vote based on the features observed in its assigned dimension. The final classification is obtained via majority voting.

This design allows the model to capture unique temporal dynamics within each channel while leveraging the ensemble’s power to model cross-channel correlations implicitly. It scales efficiently with the number of dimensions and maintains the interpretability of individual trees.

3.5 Model Optimization

To prevent overfitting, we apply **post-pruning**. Subtrees are replaced with leaf nodes if the change does not significantly increase the error rate on a validation set. This results in more compact, differentiable trees that generalize better to unseen data.

4 Experiments and Results

We evaluated MTSST on a diverse set of datasets from the UCR (univariate) and UEA (multivariate) archives. We compare the standalone TSST Decision Tree (DT) and its Random Forest (RF) extension.

4.1 Univariate Results

We tested the model on 13 univariate datasets. The TSST model showed robust performance, often effectively classifying time series with high accuracy. For example, on the *Chinatown* dataset, the RF-TSST achieved an accuracy of 97.7%, with high Cohen’s Kappa (0.941), demonstrating excellent reliability.

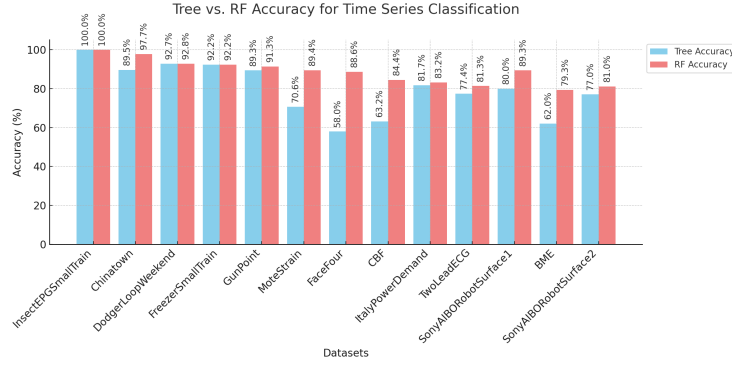


Fig. 1. Classification accuracy of TSST (DT) vs. RF-TSST on univariate datasets.

4.2 Multivariate Results

The primary focus of this paper is the multivariate extension (MTSST). We evaluated the RF-based MTSST on 6 representative datasets from the UEA archive. Table 1 summarizes the characteristics and accuracy results.

Table 1. Dataset characteristics and test accuracy for multivariate benchmarks using RF-TSST.

Dataset	No. Features		Classes	Dims	Length	ACC (RF-TSST)
	Train	Test				
Epilepsy	137	138	4	3	206	99.3%
BasicMotions	40	40	4	6	100	97.5%
SelfRegulationSCP1	268	293	2	6	896	95.0%
Cricket	108	72	12	6	1197	83.3%
ArticularyWordRecognition	275	300	25	9	144	73.0%
Heartbeat	204	205	2	61	405	73.0%

Performance Analysis. The model achieves exceptional accuracy on datasets like *Epilepsy* (99.3%) and *BasicMotions* (97.5%). Even on more complex tasks with many classes, such as *Cricket*, it maintains competitive performance. Notably, in the *Epilepsy* dataset, per-class metrics (Precision, Recall, F1) cluster near 1.0 for all classes (epilepsy, running, sawing, walking), indicating that the random forest ensemble effectively captures the distinct temporal and cross-channel patterns required for discrimination.

4.3 Interpretability and Promptness

A key advantage of MTSST is its "promptness"—the ability to make decisions early. Figure 3 illustrates a sample tree from the Epilepsy dataset. The tree shows

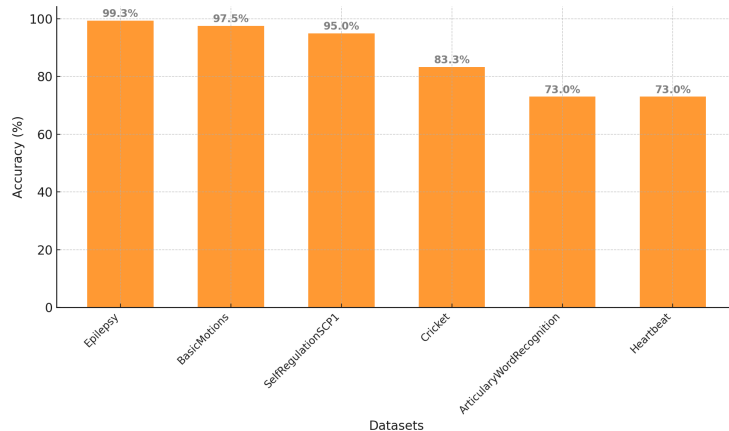


Fig. 2. Classification accuracy of RF-TSST across six multivariate datasets.

how the model splits based on short subsequences (e.g., Euclidean distance to a Witness-TS of length 21), often reaching a leaf node well before processing the full sequence length (206 points).

We analyzed the distribution of "steps" required for decision-making. In many cases, the median number of steps required to reach a classification is significantly lower than the full time series length. This confirms that MTSST effectively balances accuracy with promptness, consuming only as much data as necessary for a confident decision.

5 Conclusion

In this paper, we presented **MTSST**, a novel framework for the prompt and interpretable classification of multivariate time series. By extending the Time Series Step Tree with a dimension-wise random forest ensemble, MTSST captures complex multivariate dependencies while retaining the transparency of step-wise decision trees. Empirical results on UEA datasets demonstrate that MTSST achieves state-of-the-art accuracy (up to 99.3% on Epilepsy) and provides valuable insights into the decision-making process through its explicit Witness-TS selection and dynamic observation windows. Future work will focus on optimizing the ensemble strategy and exploring applications in real-time health monitoring systems.

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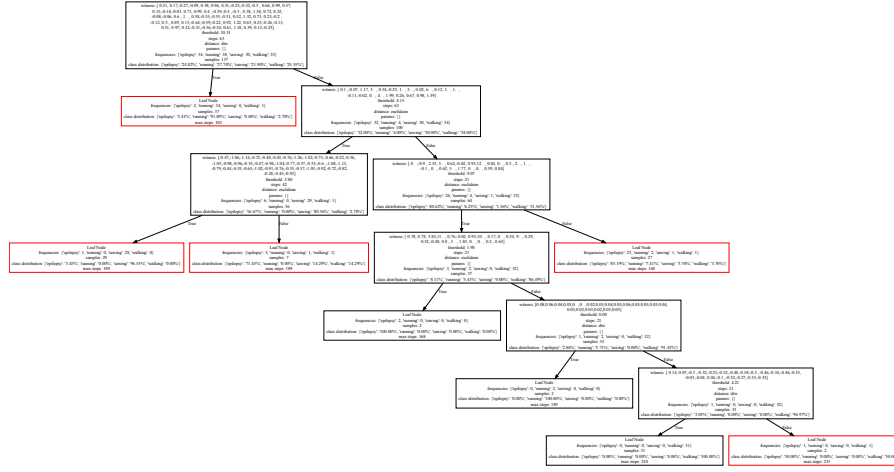


Fig. 3. Dimension-2 decision tree from the RF-TSST on the Epilepsy dataset. Nodes display the Witness-TS, threshold, and steps used.

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